

Minimum Edge Sets under Diamond Closure in Boolean and Distributive Lattices

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Abstract

We study minimum-cardinality sets of cover edges under diamond closure in Boolean and finite distributive lattices. For a Boolean lattice B_X , we classify the $|X|$ -edge sets whose diamond closure contains a maximal chain. Such a set is encoded by a loopless digraph, and chain-generation is governed by a decomposition into out-modules. The induced-minimal obstructions consist of four three-vertex digraphs together with an infinite family F_k . For a finite distributive lattice $\mathcal{J}(P)$, every full diamond-generating set has at least $|P|$ edges, and the minimum sets are characterized by incomparability symmetry and the same family F_k . Every F_k is realizable in a finite distributive lattice. For products of chains only F_2 and F_3 remain; for two chains the minimum generators are in bijection with Ferrers diagrams.

1 Introduction

Diamond completion on cover graphs arose in the study of pseudo-roots of noncommutative polynomials. Gelfand, Gelfand, Retakh, and Wilson connected sets of edges in directed graphs with rational generation of pseudo-roots [4]. Retakh and Saks developed diamond closure on cover graphs of finite modular lattices and gave structural descriptions of diamond-closed sets and their closures [7]. Their distributive case uses the representation of finite distributive lattices as lattices of order ideals [1].

This paper studies the minimum-edge regime. Two problems are considered:

- classify $|X|$ -edge subsets of $E(B_X)$ whose closure contains a maximal chain,
- classify $|P|$ -edge subsets of $E(\mathcal{J}(P))$ whose closure is $E(\mathcal{J}(P))$.

The same encoding controls both problems. A minimum Boolean edge set has one edge of each label and therefore determines a loopless digraph. Under the local in-neighborhood condition in (M), its asymmetric arcs determine a poset P_D . The closure then decomposes along out-modules. The term out-module is standard in digraph theory [8]; the use of decomposition trees also fits the broader modular-decomposition framework for graph-like structures [2].

Theorem 6 gives the common structural reduction. Section 5 classifies induced-minimal failures of that reduction. Every such failure has a cyclic normal form, and in-neighborhood monotonicity reduces the normal forms to the family F_k . Theorem 17 applies this classification to Boolean chain-generation. Theorem 21 applies it to full generation in $\mathcal{J}(P)$. The final sections specialize the result to products of chains and to the two-chain Ferrers case.

A previous paper treated full diamond-generation in Boolean lattices and obtained cographs as the minimum generators [6]. The present classification places that result inside the distributive-lattice setting; the relation is recorded after Theorem 21.

2 Setup

Let X be a finite set and let B_X be its Boolean lattice. For $i \in X$ and $A \subseteq X \setminus \{i\}$, write

$$e_i(A) : A \longrightarrow A \cup \{i\}$$

for the cover edge labelled by i . We orient all cover edges upward. For distinct $i, j \notin A$, the four edges

$$e_i(A), \quad e_j(A), \quad e_i(A \cup \{j\}), \quad e_j(A \cup \{i\})$$

form a diamond. A *diamond completion* adjoins the upper pair when the lower pair is present, or the lower pair when the upper pair is present. For an edge set S , let $\text{cl}_\diamond(S)$ denote the least edge set containing S and closed under diamond completion.

An edge set $S \subseteq E(B_X)$ is *chain-generating* if $\text{cl}_\diamond(S)$ contains a maximal chain from $\emptyset$ to X . Since a maximal chain contains one edge of each label and diamond completion never introduces a new label, every chain-generating set has at least $|X|$ edges. If $|S| = |X|$, then a chain-generating set has exactly one edge of each label, so it can be written

$$S_D = \{e_i(A_i) : i \in X\}, \quad A_i = N_D^+(i),$$

for a unique loopless digraph D on X , where

$$i \rightarrow j \iff j \in A_i.$$

We call D the *encoding digraph* of the edge set.

For a finite poset P , let $\mathcal{J}(P)$ denote the lattice of order ideals of P . A cover edge of $\mathcal{J}(P)$ has the form

$$e_p(A) : A \longrightarrow A \cup \{p\},$$

where $A \in \mathcal{J}(P)$, $p \notin A$, and every predecessor of p lies in A . We say that $S \subseteq E(\mathcal{J}(P))$ *diamond-generates* $\mathcal{J}(P)$ if $\text{cl}_\diamond(S)$ is the entire cover graph of $\mathcal{J}(P)$.

3 Diamond closure and an induced poset

We shall use the following condition on a loopless digraph D :

$$u \rightarrow v, \quad v \not\rightarrow u \implies N_D^-(u) \subseteq N_D^-(v). \quad (\text{M})$$

We refer to (M) as *in-neighborhood monotonicity*.

Assume D satisfies (M). Define a relation P_D on X by

$$x <_{P_D} y \iff y \rightarrow x \text{ and } x \not\rightarrow y. \quad (1)$$

Lemma 1 (Induced poset). *The relation P_D is a strict partial order. For every $i \in X$, both $A_i = N_D^+(i)$ and $A_i \cup \{i\}$ are order ideals of P_D .*

Proof. Suppose $u \rightarrow v$ and $v \rightarrow w$ are asymmetric arcs. Since $u \in N^-(v)$, applying (M) to $v \rightarrow w$ gives $u \rightarrow w$. If $w \rightarrow u$, then applying (M) to $u \rightarrow v$ gives $w \rightarrow v$, contradicting the asymmetry of $v \rightarrow w$. Thus asymmetric arcs are transitive and contain no directed cycle, so (1) defines a strict partial order.

Now suppose $x <_{P_D} y$ and $y \in A_t$. Then $y \rightarrow x$ is asymmetric and $t \rightarrow y$, so

$$t \in N^-(y) \subseteq N^-(x).$$

Hence $t \rightarrow x$, so $x \in A_t$. Therefore A_t is an ideal. If $x <_{P_D} t$, then $t \rightarrow x$, so every predecessor of t already lies in A_t ; hence $A_t \cup \{t\}$ is also an ideal. $\square$

For a family $\mathcal{V} \subseteq B_X$, let $\langle \mathcal{V} \rangle_{\cup, \cap}$ denote the sublattice generated by $\mathcal{V}$. If $K \subseteq B_X$ is a sublattice, let $E(K)$ denote the Boolean cover edges whose two endpoints lie in K .

The following lemma and its distributive corollary are the present-form specialization of the connected diamond-closure description of Retakh and Saks [7]. We include the proof in this notation.

Lemma 2 (Connected diamond closure). *Let T be a nonempty connected set of cover edges of B_X . Then*

$$\text{cl}_\diamond(T) = E\left(\langle V(T) \rangle_{\cup, \cap}\right). \quad (2)$$

Proof. Put $H = \text{cl}_\diamond(T)$. Then H is connected and diamond-closed. Take $U, V \in V(H)$ and a shortest walk in H from U to V . Whenever two consecutive steps use distinct labels, they may be interchanged across the corresponding Boolean diamond. Two consecutive steps with the same label and opposite signs form a removable backtrack.

No label can occur twice in a shortest walk. Indeed, choose two consecutive occurrences of a repeated label. They have opposite signs, and every intervening step has a different label. Commuting one occurrence across the intervening steps makes the two occurrences adjacent, producing a backtrack and contradicting shortestness. Hence the walk changes precisely the elements of $U \triangle V$.

Commute all downward steps before all upward steps. The resulting walk remains in H and passes through $U \cap V$. Commuting in the opposite direction gives a walk through $U \cup V$. Thus $V(H)$ is a sublattice. If two vertices of $V(H)$ differ by one element, the preceding argument gives a walk of length one between them, so their cover edge lies in H . Hence

$$H = E(V(H)).$$

Let $K = \langle V(T) \rangle_{\cup, \cap}$. Since $V(H)$ is a sublattice containing $V(T)$, we have $K \subseteq V(H)$. Conversely, $E(K)$ is diamond-closed and contains T , so minimality of diamond closure gives $H \subseteq E(K)$. Therefore $V(H) = K$, proving (2). $\square$

Corollary 3 (Connected diamond closure in distributive lattices). *Let P be a finite poset and let T be a nonempty connected set of cover edges of $\mathcal{J}(P)$. Then*

$$\text{cl}_\diamond(T) = E\left(\langle V(T) \rangle_{\cup, \cap}\right),$$

where the closure and cover edges are taken in $\mathcal{J}(P)$.

Proof. Regard $\mathcal{J}(P)$ as a sublattice of B_P . Starting from cover edges of $\mathcal{J}(P)$, every Boolean diamond completion remains in $\mathcal{J}(P)$: the fourth vertex of the square is obtained from ideals by union or intersection. Thus the diamond closure computed in B_P agrees with the closure computed in $\mathcal{J}(P)$. Lemma 2 applies, and the sublattice generated by $V(T)$ remains inside $\mathcal{J}(P)$. $\square$

4 Out-module decomposition

Continue to assume that D satisfies (M). The chosen edge endpoints generate the entire induced distributive lattice.

Lemma 4 (Endpoint generation).

$$\langle A_i, A_i \cup \{i\} : i \in X \rangle_{\cup, \cap} = \mathcal{J}(P_D). \quad (3)$$

Proof. By Lemma 1, every chosen endpoint is an ideal of P_D , so the generated lattice is contained in $\mathcal{J}(P_D)$.

Fix $y \in X$ and intersect all chosen endpoints containing y . Each such endpoint is an ideal and therefore contains $\downarrow y$. Let $x \not\leq_{P_D} y$. If $x \notin A_y$, then $A_y \cup \{y\}$ contains y and excludes x . If $x \in A_y$, then $y \rightarrow x$; since $x \not\leq_{P_D} y$, the arc is not asymmetric, so $x \rightarrow y$ as well. Then A_x contains y and excludes x . Hence the intersection is exactly $\downarrow y$.

Also

$$\bigcap_{i \in X} A_i = \emptyset,$$

because $i \notin A_i$ for every i . Every ideal is a union of principal ideals, so the chosen endpoints generate all of $\mathcal{J}(P_D)$. $\square$

Following standard digraph terminology [8], a nonempty set $C \subseteq X$ is an *out-module* of D if

$$N_D^+(x) \setminus C = N_D^+(y) \setminus C \quad (x, y \in C).$$

Write the common external out-neighborhood as R_C . Since D is loopless,

$$R_C = \bigcap_{i \in C} A_i, \tag{4}$$

so R_C is an ideal of P_D . Define

$$K_C = \{R_C \cup I : I \in \mathcal{J}(P_D[C])\}. \tag{5}$$

This is an embedded copy of $\mathcal{J}(P_D[C])$ inside $\mathcal{J}(P_D)$. Indeed, if $x \in C$, $y \notin C$, and $y <_{P_D} x$, then $x \rightarrow y$, so $y \in A_x \setminus C = R_C$. Applying Lemma 4 to $D[C]$ and translating by R_C gives

$$K_C = \langle A_i, A_i \cup \{i\} : i \in C \rangle_{\cup, \cap}. \tag{6}$$

Lemma 5 (Intersection criterion). *If A, B are disjoint nonempty out-modules, then*

$$K_A \cap K_B \neq \emptyset \iff A \cup B \text{ is an out-module.} \tag{7}$$

Proof. Suppose first that $W \in K_A \cap K_B$. Outside $A \cup B$, membership in every member of K_A is fixed by R_A , while membership in every member of K_B is fixed by R_B . Hence

$$R_A \setminus B = R_B \setminus A.$$

Thus the vertices of $A \cup B$ have the same out-neighbors outside $A \cup B$, so $A \cup B$ is an out-module.

Conversely, suppose $A \cup B$ is an out-module, with common external out-neighborhood R . Then

$$R_A = R \cup (R_A \cap B), \quad R_B = R \cup (R_B \cap A).$$

Since R_A and R_B are ideals,

$$R_B \cap A \in \mathcal{J}(P_D[A]), \quad R_A \cap B \in \mathcal{J}(P_D[B]).$$

Therefore

$$R \cup (R_B \cap A) \cup (R_A \cap B)$$

belongs to both K_A and K_B . $\square$

Call a digraph D *out-module decomposable* if the singleton partition of $V(D)$ can be reduced to $\{V(D)\}$ by repeatedly merging two current blocks whose union is an out-module. Equivalently, there is a rooted binary tree with leaf set $V(D)$ such that the leaves below every node form an out-module.

Theorem 6 (Out-module decomposition). *Assume D satisfies (M). Start with the singleton partition and repeatedly merge two current blocks whenever their union is an out-module. Let Π be any terminal partition. Then*

$$\text{cl}_\diamond(S_D) = \bigsqcup_{C \in \Pi} E(K_C). \quad (8)$$

The partition Π is independent of the sequence of merges. Consequently,

$$S_D \text{ is chain-generating} \iff D \text{ is out-module decomposable.} \quad (9)$$

When these conditions hold,

$$\text{cl}_\diamond(S_D) = E(\mathcal{J}(P_D)). \quad (10)$$

Proof. For every current block C , maintain the invariant

$$\text{cl}_\diamond(\{e_i(A_i) : i \in C\}) = E(K_C). \quad (11)$$

For a singleton, this is immediate. Suppose current blocks A, B are merged. By Lemma 5, $K_A \cap K_B \neq \emptyset$, so $E(K_A) \cup E(K_B)$ is connected. By Lemma 2, its diamond closure is the cover graph of the sublattice generated by $K_A \cup K_B$. By (6), that sublattice is $K_{A \cup B}$. Thus (11) is preserved.

At termination, Lemma 5 shows that K_C and $K_{C'}$ are vertex-disjoint for distinct terminal blocks. Each $E(K_C)$ is diamond-closed. Their union is also diamond-closed, because two edges that trigger a diamond share an endpoint and therefore cannot lie in two vertex-disjoint members. This proves (8).

Each $E(K_C)$ is connected, and its set of edge labels is exactly C . Hence the connected components in (8) recover Π , proving that the terminal partition is intrinsic.

If $\Pi = \{X\}$, then $R_X = \emptyset$ and $K_X = \mathcal{J}(P_D)$, which gives (10). In particular, a linear extension of P_D gives a maximal Boolean chain inside $\mathcal{J}(P_D)$, so S_D is chain-generating. If $|\Pi| > 1$, each connected component $E(K_C)$ uses only labels in the proper subset C . A maximal Boolean chain is connected and uses every label of X , so no component can contain one. Thus S_D is not chain-generating. $\square$

5 Minimal obstructions to out-module decomposition

Theorem 6 reduces the generation questions to a hereditary digraph property. We classify its induced-minimal obstructions.

Out-module decomposability is hereditary under induced subdigraphs: restrict a decomposition tree to an induced vertex set and suppress nodes with only one surviving child. We also use the following operation. For a vertex v , *complement the in-neighborhood at v* by replacing $N^-(v)$ with $V(D) \setminus (N^-(v) \cup \{v\})$, leaving all other arcs unchanged. Complementing the in-neighborhoods at any collection of vertices preserves every out-module, because for a vertex outside a prospective out-module all incidences from the module are toggled simultaneously. Hence it preserves out-module decomposability and induced minimality of nondecomposability.

Call D *induced-minimal nondecomposable* if D is not out-module decomposable but every proper nonempty induced subdigraph is.

Lemma 7. *Every strongly connected component of an out-module decomposable digraph is an out-module.*

Proof. Induct on the number of vertices. Let A, B be the two children of the root of an out-module decomposition tree. There are sets $U \subseteq A$ and $V \subseteq B$ such that

$$N^+(b) \cap A = U \quad (b \in B), \quad N^+(a) \cap B = V \quad (a \in A). \quad (12)$$

If $U = \emptyset$ or $V = \emptyset$, no strongly connected component meets both A and B . Every component lies in one child, where induction applies; an out-module of an out-module is an out-module of the whole digraph.

Suppose $U, V \neq \emptyset$. Every $u \in U$ and $v \in V$ form a directed 2-cycle. Hence $U \cup V$ lies in one strongly connected component K . Every vertex reachable from $U \cup V$ can return to it: a vertex in A points to every vertex of V , and a vertex in B points to every vertex of U . Thus K is exactly the set reachable from $U \cup V$. It has no outgoing arc to its complement, so it is an out-module. Every other strongly connected component lies in one child and is handled inductively. $\square$

Lemma 8. *If every strongly connected component of D is an out-module and induces an out-module decomposable digraph, then D is out-module decomposable.*

Proof. Decompose each strongly connected component internally. Order the components in reverse topological order in the condensation digraph, with sinks first. Every prefix union has no outgoing arc to its complement and is therefore an out-module. Merge the components successively in that order. $\square$

Fix an induced-minimal nondecomposable digraph D and a vertex r . For each $j \neq r$ with $r \rightarrow j$, complement the in-neighborhood at j . Let the resulting digraph be H . Then r is a sink and H is again induced-minimal nondecomposable. Put

$$G = H - r$$

and color its vertices by

$$\chi(x) = \mathbf{1}_{\{x \rightarrow_H r\}}. \quad (13)$$

Lemma 9. *The digraph G is out-module decomposable and strongly connected, both colors occur, and no proper induced strongly connected subdigraph of G contains both colors.*

Proof. By minimality, G is out-module decomposable. Hence every strongly connected component of G is an out-module of G by Lemma 7.

Since r is a sink, the strongly connected components of H are $\{r\}$ together with those of G . If every component of G were also an out-module of H , then Lemma 8 would make H out-module decomposable, a contradiction. Thus some strongly connected component C of G is not an out-module of H . Its vertices have identical out-neighborhoods outside C within G , so their disagreement in H must be at the vertex r . Hence both colors occur in C .

Then $H[C \cup \{r\}]$ is not out-module decomposable, because C is a strongly connected component that is not an out-module. Minimality forces $C \cup \{r\} = V(H)$, so $C = G$. The same argument applies to any proper induced strongly connected $Y \subsetneq G$ containing both colors: $H[Y \cup \{r\}]$ would be a proper induced nondecomposable subdigraph. Thus none exists. $\square$

Lemma 10 (Singleton-color lemma). *Let G be an out-module decomposable strongly connected digraph on at least three vertices. Color $V(G)$ with two colors, both present. If no proper induced strongly connected subdigraph contains both colors, then one color occurs at exactly one vertex.*

Proof. Let A, B be the root clusters of an out-module decomposition tree and define $U \subseteq A, V \subseteq B$ by (12). Strong connectivity gives $U, V \neq \emptyset$. Every $u \in U$ and $v \in V$ form a directed 2-cycle. Since $|V(G)| \geq 3$, the induced subdigraph on $\{u, v\}$ is proper and strongly connected, so u and v have the same color. Hence all of $U \cup V$ have one color, say 0.

Choose a vertex of color 1, interchanging A, B if necessary so that one lies in A . There is a directed path contained in A from U to a color-1 vertex: take a path from B to such a vertex and retain the suffix after its last entrance into A . Choose a shortest such path

$$x_0, x_1, \dots, x_t, \quad x_0 \in U.$$

Then $x_0, \dots, x_{t-1}$ have color 0, while x_t has color 1. Fix $v \in V$. The induced subdigraph on

$$Y = \{v, x_0, \dots, x_t\}$$

is strongly connected: $v \rightarrow x_0$, the displayed path reaches every x_i , and every $x_i \rightarrow v$. It contains both colors, so by hypothesis $Y = V(G)$. Therefore x_t is the unique vertex of color 1. $\square$

Corollary 11. *Let D be induced-minimal nondecomposable on $n \geq 4$ vertices. For every $v \in V(D)$ there are unique*

$$t_v \in \{0, 1\}, \quad f(v) \in V(D) \setminus \{v\},$$

such that, for $u \neq v$,

$$\mathbf{1}_{\{u \rightarrow v\}} = \begin{cases} 1 - t_v, & u = f(v), \\ t_v, & u \notin \{v, f(v)\}. \end{cases} \quad (14)$$

Thus, for each v , adjacency into v is constant on $V(D) \setminus \{v, f(v)\}$ and takes the opposite value at $f(v)$.

Proof. Apply Lemmas 9 and 10 with $r = v$. The in-neighborhood of v was not complemented, so the coloring in (13) is exactly the original adjacency into v . One color therefore occurs once among the $n - 1$ vertices other than v . Since $n - 2 \geq 2$, the majority value t_v and exceptional vertex $f(v)$ are unique. $\square$

Lemma 12 (Spanning exception cycle). *For an induced-minimal nondecomposable digraph on $n \geq 4$ vertices, the functional digraph of the map $v \mapsto f(v)$ is a single directed cycle containing all vertices.*

Proof. There are no fixed points. Suppose f has a directed cycle Z of length at least three. For any distinct $p, q \in Z$, at least one predecessor of p or q on the cycle lies outside $\{p, q\}$. Thus there is $v \in Z \setminus \{p, q\}$ with $f(v) \in \{p, q\}$. By (14), exactly one of p, q has the majority adjacency value into v , so $\{p, q\}$ is not an out-module of $D[Z]$. Hence no two-vertex subset of Z is an out-module. An out-module decomposition must begin by merging two singleton blocks, so $D[Z]$ is not decomposable. Minimality forces $Z = V(D)$.

It remains to exclude the case in which every directed cycle of f has length two. Suppose some vertex lies outside those cycles. Choose a vertex p immediately preceding a two-cycle along a functional path. Then there are distinct p, q, s with

$$f(p) = q, \quad f(q) = s, \quad f(s) = q.$$

Inside $\{p, q, s\}$, adjacency into s distinguishes p from q , adjacency into q distinguishes p from s , and adjacency into p distinguishes q from s . Thus no pair is an out-module, so this proper three-vertex induced subdigraph is nondecomposable, contradicting minimality.

Therefore every vertex lies in a two-cycle. Each such pair is an out-module: for any vertex outside the pair, its exceptional vertex belongs to its own two-cycle and therefore lies outside the pair. The same argument shows that every union of whole two-cycles is an out-module. Merging each pair and then the resulting blocks gives an out-module decomposition, again a contradiction. Hence f is one spanning cycle. $\square$

Theorem 13 (Cyclic normal form). *Every induced-minimal nondecomposable digraph on $n \geq 4$ vertices has a cyclic ordering*

$$v_0, v_1, \dots, v_{n-1}$$

and bits $t_0, \dots, t_{n-1}$ such that

$$f(v_i) = v_{i-1}$$

(indices modulo n) and

$$\mathbf{1}_{\{v_j \rightarrow v_i\}} = \begin{cases} 1 - t_i, & j = i - 1, \\ t_i, & j \notin \{i, i - 1\}. \end{cases} \quad (15)$$

Conversely, every digraph of the form (15) is induced-minimal nondecomposable.

Proof. The forward direction is Corollary 11 and Lemma 12.

Conversely, let D have the form (15). For any distinct p, q , some $v \notin \{p, q\}$ has $f(v) \in \{p, q\}$, because f is a spanning cycle of length at least four. Hence adjacency into v distinguishes p and q . No pair is an out-module, so no first merge is possible.

Let $Y \subsetneq V(D)$ be nonempty. The retained functional arcs

$$v \longrightarrow f(v), \quad v, f(v) \in Y,$$

form a disjoint union of directed paths. Choose an ordering of Y in which v precedes $f(v)$ whenever both are retained. Every prefix C of this ordering is an out-module of $D[Y]$. Indeed, take $w \in Y \setminus C$. If $f(w) \in C$, the ordering condition would put w before $f(w)$ and hence force $w \in C$, a contradiction. Thus $f(w) \notin C$, and (15) shows that every vertex of C has the same adjacency value into w . Therefore C is an out-module. The successive prefixes yield an out-module decomposition of $D[Y]$. Hence D is induced-minimal nondecomposable. $\square$

For $k \geq 2$, define F_k on

$$C = \{c_0, \dots, c_{k-1}\}, \quad U = \{u_0, \dots, u_{k-1}\}$$

by

$$N^+(u_i) = C \setminus \{c_{i+1}\}, \quad N^+(c_i) = (C \setminus \{c_i\}) \cup \{u_i\}, \quad (16)$$

with indices modulo k . Thus C induces a complete symmetric digraph, U is independent, u_i and c_i form a directed 2-cycle, u_i and c_{i+1} are nonadjacent, and every remaining cross pair has the single arc $u_i \rightarrow c_j$. For $k = 2$, F_2 is the bidirected P_4 .

Figure 1 shows the first two members of the family.

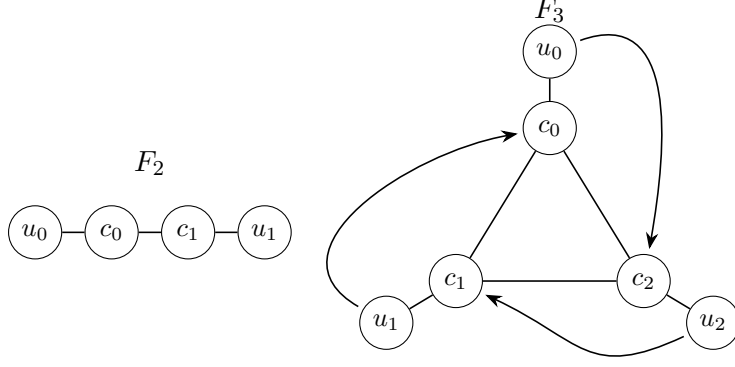


Figure 1: The obstructions F_2 and F_3 . Plain edges denote directed 2-cycles. Arrows denote asymmetric arcs.

Lemma 14. *Every digraph satisfying (M) on at most three vertices is out-module decomposable.*

Proof. The statement is immediate for one or two vertices. Suppose a three-vertex digraph satisfying (M) were not out-module decomposable. Then no pair is an out-module. Consequently, for every vertex v , the other two vertices have different adjacency values into v , so every vertex has indegree one. If $u \rightarrow v$ were asymmetric, (M) would give $N^-(u) \subseteq N^-(v)$. The inclusion is strict because $u \in N^-(v) \setminus N^-(u)$, contradicting equality of the two indegrees. Thus every arc belongs to a directed 2-cycle. The underlying simple graph would have three vertices each of degree one, which is impossible. $\square$

Theorem 15 (Minimal obstructions under in-neighborhood monotonicity). *The induced-minimal nondecomposable digraphs satisfying (M) are exactly*

$$F_k, \quad k \geq 2. \quad (17)$$

Proof. Let D be induced-minimal nondecomposable and satisfy (M). By Lemma 14, $|V(D)| \geq 4$, so Theorem 13 applies. Take consecutive vertices

$$w = v_{i-2}, \quad u = v_{i-1}, \quad v = v_i.$$

If $t_{i-1} = t_i = 0$, then (15) gives

$$u \rightarrow v, \quad v \not\rightarrow u, \quad w \rightarrow u, \quad w \not\rightarrow v,$$

contradicting (M). If $t_{i-1} = t_i = 1$, then

$$v \rightarrow u, \quad u \not\rightarrow v, \quad w \rightarrow v, \quad w \not\rightarrow u,$$

again contradicting (M). Hence

$$t_{i-1} \neq t_i \quad \text{for every } i.$$

Thus the cycle has even length $2k$. Choose the cyclic origin so that

$$v_{2i} = c_i, \quad t_{2i} = 1, \quad v_{2i+1} = u_i, \quad t_{2i+1} = 0.$$

Then $f(u_i) = c_i$ and $f(c_{i+1}) = u_i$. Reading (15) gives exactly (16). Thus every minimal obstruction satisfying (M) is some F_k .

Conversely, F_k has the cyclic normal form (15) with alternating bits, so Theorem 13 gives induced minimality. Its asymmetric arcs are precisely

$$u_i \rightarrow c_j, \quad j \notin \{i, i+1\}.$$

Moreover $N^-(u_i) = \{c_i\}$. For every such asymmetric arc, $c_i \rightarrow c_j$ because C is complete symmetric. Hence

$$N^-(u_i) = \{c_i\} \subseteq N^-(c_j),$$

so (M) holds. $\square$

6 Minimum chain-generating sets in Boolean lattices

We first apply the obstruction theorem to the maximal-chain problem in the Boolean lattice.

Chain-generation is hereditary under induced subdigraphs of the encoding digraph. Indeed, if $Y \subseteq X$, projection by $A \mapsto A \cap Y$ sends S_D to $S_{D[Y]}$ and sends every diamond completion to either the corresponding completion in B_Y or a collapsed edge. A maximal chain in B_X projects, after repetitions are removed, to a maximal chain in B_Y .

There are four relevant three-vertex digraphs. With all unlisted arcs absent, define

	$N^+(a)$	$N^+(b)$	$N^+(c)$
Q_1	$\{b\}$	$\{c\}$	$\emptyset$
Q_2	$\{b, c\}$	$\{a\}$	$\emptyset$
Q_3	$\{b\}$	$\{c\}$	$\{a\}$
Q_4	$\{c\}$	$\{a\}$	$\{a, b\}$.

(18)

Figure 2 shows these four digraphs.

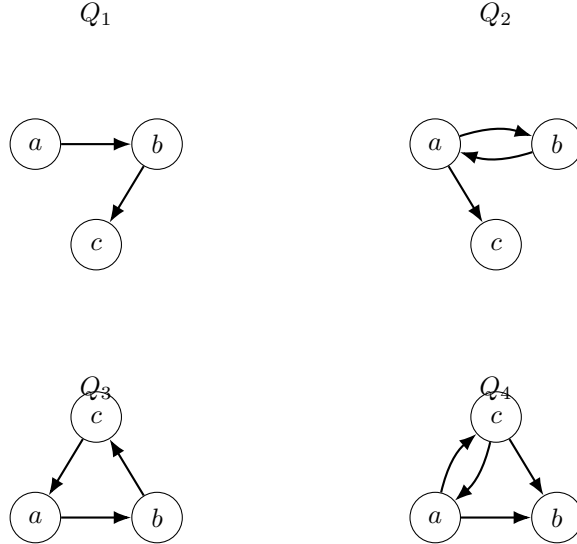


Figure 2: The three-vertex obstructions Q_1, Q_2, Q_3, Q_4 for Boolean chain-generation.

Lemma 16. *A loopless digraph satisfies (M) if and only if it contains none of Q_1, Q_2, Q_3, Q_4 as an induced subdigraph.*

Proof. Suppose (M) fails. Then there are distinct u, v, w with

$$u \rightarrow v, \quad v \not\rightarrow u, \quad w \rightarrow u, \quad w \not\rightarrow v.$$

The only undetermined arcs on these three vertices are $u \rightarrow w$ and $v \rightarrow w$. Their four possibilities give respectively Q_1, Q_3, Q_2, Q_4 . Conversely, each of the four digraphs in (18) contains such a triple. $\square$

For each Q_i , the three lower endpoints of S_{Q_i} are pairwise distinct and the three upper endpoints are pairwise distinct. Hence S_{Q_i} is diamond-closed. Its three edges do not form a maximal chain, so S_{Q_i} is not chain-generating. Every loopless digraph on at most two vertices encodes a chain-generating edge set. Therefore the Q_i are induced-minimal non-chain-generating encodings.

Theorem 17 (Boolean chain-generation classification). *Let D be a finite loopless digraph on X . Then the $|X|$ -edge set*

$$S_D = \{e_i(N_D^+(i)) : i \in X\}$$

is chain-generating if and only if D contains no induced Q_1, Q_2, Q_3, Q_4 and no induced F_k for $k \geq 2$. Consequently, the induced-minimal non-chain-generating encoding digraphs are precisely

$$Q_1, Q_2, Q_3, Q_4, \quad F_k \quad (k \geq 2).$$

Whenever D contains none of these obstructions,

$$\text{cl}_\diamond(S_D) = E(\mathcal{J}(P_D)).$$

Proof. If S_D is chain-generating, heredity excludes every Q_i . By Lemma 16, D therefore satisfies (M). Theorem 6 then shows that D is out-module decomposable. By Theorem 15, it contains no induced F_k .

Conversely, suppose D contains none of the Q_i and none of the F_k . Lemma 16 gives (M). If D were not out-module decomposable, an induced-minimal nondecomposable subdigraph would still satisfy (M); by Theorem 15, it would be some F_k , a contradiction. Thus D is out-module decomposable, and Theorem 6 gives both chain-generation and the displayed closure formula.

The Q_i are induced-minimal non-chain-generating as observed above. Each F_k is induced-minimal nondecomposable and satisfies (M); Theorem 6 therefore makes S_{F_k} induced-minimal non-chain-generating. There are no further minimal obstructions by the equivalence just proved. $\square$

7 Minimum diamond-generating sets in distributive lattices

We now prescribe the poset and ask for full diamond-generation of its ideal lattice.

Let P be a finite poset. For $p \in P$, write

$$\text{Pred}(p) = \{q \in P : q < p\}.$$

Every cover edge of $\mathcal{J}(P)$ has a unique label p , namely the element added by that edge. Since diamond completion preserves labels, every edge set that diamond-generates $\mathcal{J}(P)$ has at least $|P|$ edges.

Lemma 18. *For every finite poset P , the minimum cardinality of an edge set that diamond-generates $\mathcal{J}(P)$ is $|P|$.*

Proof. Only the upper bound remains. Take

$$S_0 = \{\text{Pred}(p) \longrightarrow \text{Pred}(p) \cup \{p\} : p \in P\}.$$

We prove by induction on $|A|$ that every cover edge

$$A \longrightarrow A \cup \{p\}$$

belongs to $\text{cl}_\diamond(S_0)$. If $A = \text{Pred}(p)$, the edge lies in S_0 . Otherwise choose q maximal in $A \setminus \text{Pred}(p)$. Then q is maximal in A : if $q < z \in A$, then $z < p$ would imply $q < p$, while $z \not\leq p$ would place z in $A \setminus \text{Pred}(p)$ above q . Furthermore $q \parallel p$, because $q \not\leq p$ by choice while $p < q$ would force $p \in A$. Put $B = A \setminus \{q\}$. Then B is an ideal, and both

$$B \longrightarrow B \cup \{p\}, \quad B \longrightarrow B \cup \{q\} = A$$

are cover edges. Their lower endpoints have smaller cardinality, so induction supplies both. Completing their diamond gives $A \rightarrow A \cup \{p\}$. Thus S_0 diamond-generates $\mathcal{J}(P)$. $\square$

Accordingly, let

$$S = \{A_p \longrightarrow A_p \cup \{p\} : p \in P\} \quad (19)$$

contain one cover edge of each label, and define its encoding digraph D_S on P by

$$p \rightarrow q \iff q \in A_p. \quad (20)$$

For $Q \subseteq P$, projection by

$$\pi_Q : \mathcal{J}(P) \longrightarrow \mathcal{J}(P[Q]), \quad I \longmapsto I \cap Q$$

is surjective. A diamond projects either to a diamond, to a retained edge counted twice, or to a point, and every cover edge of $\mathcal{J}(P[Q])$ lifts to a cover edge of $\mathcal{J}(P)$. Hence diamond-generation is inherited by restriction to Q .

Lemma 19 (Incomparability symmetry). *If S diamond-generates $\mathcal{J}(P)$, then for every incomparable pair $p \parallel q$,*

$$p \rightarrow q \iff q \rightarrow p. \quad (C)$$

For comparable elements,

$$q < p \implies p \rightarrow q, \quad q \not\leq p. \quad (21)$$

Consequently, whenever (C) holds,

$$p \rightarrow q \text{ is asymmetric} \iff q < p. \quad (22)$$

Proof. Restrict to $\{p, q\}$ with $p \parallel q$. The induced lattice is a Boolean square. The selected p -edge lies on the lower or upper side according as $q \notin A_p$ or $q \in A_p$, and similarly for the q -edge. Two such edges diamond-generate the square exactly when they are the lower pair or the upper pair. Therefore $q \in A_p$ if and only if $p \in A_q$, which is (C).

If $q < p$, every lower endpoint A_p contains q , so $p \rightarrow q$. On the other hand, $p \notin A_q$, because A_q is an ideal excluding q . This proves (21), and (22) follows from (C). $\square$

Lemma 20. *If (C) holds, then D_S satisfies (M), and the induced poset P_{D_S} is exactly P .*

Proof. By (22), the asymmetric arcs of D_S are precisely the reversals of the strict order relations of P , so $P_{D_S} = P$. Now suppose $p \rightarrow q$ and $q \not\rightarrow p$. Then $q < p$. If $x \rightarrow p$, then $p \in A_x$. Since A_x is an ideal and $q < p$, we have $q \in A_x$, so $x \rightarrow q$. Hence $N^-(p) \subseteq N^-(q)$, which is (M). $\square$

Starting from cover edges of $\mathcal{J}(P)$, every diamond completion taken in the ambient Boolean lattice B_P remains inside $\mathcal{J}(P)$, because the fourth vertex is obtained by a union or intersection of ideals. Therefore the Boolean diamond closure and the diamond closure inside $\mathcal{J}(P)$ agree on such an edge set.

Theorem 21 (Distributive-lattice classification). *Let P be a finite poset and let S be as in (19). The following are equivalent:*

- (i) S diamond-generates $\mathcal{J}(P)$;
- (ii) D_S satisfies incomparability symmetry (C) and is out-module decomposable;
- (iii) D_S satisfies incomparability symmetry (C) and contains no induced F_k for any $k \geq 2$.

Proof. If S diamond-generates $\mathcal{J}(P)$, Lemma 19 gives (C). Under (C), Lemma 20 gives (M) and $P_{D_S} = P$. Theorem 6 therefore yields

$$S \text{ diamond-generates } \mathcal{J}(P) \iff D_S \text{ is out-module decomposable,}$$

proving the equivalence of (i) and (ii).

Under the same monotonicity condition, Theorem 15 gives

$$D_S \text{ is out-module decomposable} \iff D_S \text{ contains no induced } F_k,$$

which proves the equivalence with (iii). $\square$

When P is an antichain, incomparability symmetry makes D_S symmetric. Among the obstructions F_k , only F_2 is symmetric, and F_2 is the bidirected four-vertex path. Thus Theorem 21 reduces to the P_4 -free characterization of cographs [3] and recovers the Boolean full-generation classification of [6].

The incomparability-symmetry hypothesis cannot be omitted. For the two-element antichain $P = \{p, q\}$, the two edges

$$\emptyset \longrightarrow \{p\}, \quad \{p\} \longrightarrow \{p, q\}$$

have an out-module decomposable two-vertex encoding digraph, but they form a diamond-closed directed path and do not diamond-generate the Boolean square.

8 Realizability and width

Theorem 22. *Every F_k , $k \geq 2$, occurs as the encoding digraph of an edge set containing exactly one cover edge of each label in a finite distributive lattice. Thus the edge set has the minimum cardinality that any diamond-generating set could have. For $k \geq 3$, it can be realized by a height-two poset.*

Proof. Let P_k have vertex set $C \cup U$, where

$$C = \{c_0, \dots, c_{k-1}\}, \quad U = \{u_0, \dots, u_{k-1}\},$$

and define

$$c_j < u_i \iff j \notin \{i, i+1\}, \tag{23}$$

with no other strict comparisons. For $k \geq 3$, this is a height-two poset; for $k = 2$, it is a four-element antichain.

Choose

$$A_{u_i} = C \setminus \{c_{i+1}\}, \quad A_{c_i} = (C \setminus \{c_i\}) \cup \{u_i\}. \quad (24)$$

These are valid lower endpoints of cover edges in $\mathcal{J}(P_k)$. The first contains every predecessor of u_i . For the second, every predecessor of the included element u_i lies in $C \setminus \{c_i, c_{i+1}\} \subseteq C \setminus \{c_i\}$. The resulting encoding digraph is exactly F_k , and its asymmetric arcs are precisely the reversals of the strict comparisons in P_k .

Explicitly, the selected edges are

$$C \setminus \{c_{i+1}\} \longrightarrow (C \setminus \{c_{i+1}\}) \cup \{u_i\},$$

and

$$(C \setminus \{c_i\}) \cup \{u_i\} \longrightarrow C \cup \{u_i\}.$$

All lower endpoints are distinct and all upper endpoints are distinct, so no diamond completion can begin. Hence $\text{cl}_\diamond(S) = S$. By Theorem 15, every proper induced restriction of the encoding digraph is out-module decomposable, and by Theorem 21 the corresponding restricted edge set diamond-generates its induced distributive lattice. $\square$

Corollary 23. *Even for height-two posets, minimum diamond-generation has no characterization by forbidden induced encoding digraphs of bounded order.*

Proof. The induced-minimal obstructions F_k have unbounded order $2k$ and are all realizable by height-two posets for $k \geq 3$. $\square$

For $k = 2$, the poset P_{F_2} is a four-element antichain. For $k \geq 3$, P_{F_k} has width exactly k : C is an antichain of size k , while the relations

$$c_{i+2} < u_i$$

partition its $2k$ vertices into k two-element chains.

Corollary 24. *Let P be a finite poset and let S contain one cover edge of each label.*

- (a) *If $\text{width}(P) \leq 2$, then S diamond-generates $\mathcal{J}(P)$ if and only if D_S satisfies incomparability symmetry (C).*
- (b) *If $\text{width}(P) = 3$, then after imposing (C), the only possible induced-minimal obstruction is F_3 .*

Proof. An induced F_2 requires an antichain of size four. For $k \geq 3$, an induced F_k requires width at least k . The result follows from Theorem 21. $\square$

9 Products of chains

For products of chains, the underlying poset is a disjoint union of chains, and the general obstruction family contracts to two cases. This setting also includes divisor lattices.

If

$$N = \prod_{i=1}^r p_i^{a_i},$$

then the divisor lattice of N is canonically

$$L = \prod_{i=1}^r [0, a_i],$$

where coordinate i records the exponent of p_i ; the label (i, h) corresponds to multiplication by p_i as that exponent increases from h to $h + 1$. Let

$$T = \{(i, h) : 1 \leq i \leq r, 0 \leq h < a_i\}.$$

Then $L \cong \mathcal{J}(P)$, where P is the disjoint union of the chains

$$(i, 0) < (i, 1) < \cdots < (i, a_i - 1).$$

Hence the minimum size of a diamond-generating edge set is

$$|P| = \sum_{i=1}^r a_i.$$

For the divisor lattice this is $\Omega(N)$.

For an edge set of this minimum size, let x_t be the lower endpoint of the selected edge with label $t = (i, h)$, and define

$$A_t = \{(j, \ell) \in T : \ell < x_t^j\}. \quad (25)$$

Then the encoding digraph is given by

$$t \rightarrow s \iff s \in A_t.$$

Within one coordinate chain, all arcs are forced by the chain order. For labels (i, h) and (j, ℓ) in distinct chains, incomparability symmetry (C) becomes

$$x_{(i,h)}^j > \ell \iff x_{(j,\ell)}^i > h. \quad (26)$$

The nested-neighborhood form is the standard Ferrers structure [5]. We call (26) the pairwise Ferrers consistency conditions.

Theorem 25 (Products of chains). *A minimum-size edge set in*

$$L = \prod_{i=1}^r [0, a_i]$$

diamond-generates the entire cover graph if and only if

- (i) *all pairwise Ferrers consistency conditions (26) hold;*
- (ii) *its encoding digraph contains no induced F_2 ;*
- (iii) *its encoding digraph contains no induced F_3 .*

No F_k with $k \geq 4$ can occur after (26) holds.

Proof. By Theorem 21, after (26) the only possible obstructions are the F_k . Suppose an induced F_k occurs with $k \geq 4$. In F_k , each u_i has asymmetric arcs to at least two distinct vertices of C . By (22), those vertices lie below u_i in P , hence in the same coordinate chain as u_i and are therefore comparable. But any two vertices of C form a directed 2-cycle in F_k , so by (22) they are incomparable, a contradiction. Thus only F_2 and F_3 remain, and either is an obstruction by Theorem 21.

In F_2 , every pair is either joined in both directions or nonadjacent, so its four labels must lie in four distinct coordinate chains. In F_3 , the asymmetric pairs are precisely

$$u_i \rightarrow c_{i+2} \quad (i \bmod 3),$$

so the six labels occupy exactly three coordinate chains, two labels in each. $\square$

Corollary 26. *For a product of three chains, a minimum-size edge set diamond-generates if and only if it satisfies all pairwise Ferrers consistency conditions and contains no induced F_3 . If one of the three chains has length one, pairwise Ferrers consistency alone is sufficient.*

Proof. An F_2 requires four distinct coordinate chains, and an F_3 requires two labels in each of three distinct coordinate chains. $\square$

10 The two-chain Ferrers case

Let

$$L = [0, m] \times [0, n].$$

Write the selected horizontal edges as

$$H_k : (k, a_k) \longrightarrow (k+1, a_k), \quad 0 \leq k < m,$$

and the selected vertical edges as

$$V_\ell : (b_\ell, \ell) \longrightarrow (b_\ell, \ell+1), \quad 0 \leq \ell < n.$$

Condition (26) becomes

$$a_k > \ell \iff b_\ell > k. \tag{27}$$

Hence

$$\{k : a_k > \ell\} = \{0, \dots, b_\ell - 1\},$$

so

$$a_0 \geq a_1 \geq \dots \geq a_{m-1}, \quad b_\ell = |\{k : a_k > \ell\}|. \tag{28}$$

Thus the selected edges form the southeast boundary of a Ferrers diagram in an $m \times n$ rectangle. Conversely, every such boundary satisfies (27), and no F_2 or F_3 can occur because only two coordinate chains are present.

Corollary 27. *Minimum diamond-generating edge sets of $[0, m] \times [0, n]$ are in bijection with Ferrers diagrams in an $m \times n$ rectangle. In particular,*

$$\#\{\text{minimum diamond-generating edge sets}\} = \binom{m+n}{m}.$$

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